Gao 2021 breast cancer — analysis summary

GO / NO-GO GATES

Gate 1 — mixing & continuity → NO-GO

Breast malignant cells form a per-patient archipelago. After best-effort Harmony (θ=10) the per-neighbourhood iLISI reaches only 1.6 of a 3.8 ceiling (42%), same-sample kNN stays 2.9× the mixed baseline, and 4/5 tumours are self-segregated (3.5–5.1×). Not a sample-integrated manifold.

Gate 2 — module geography → N/A

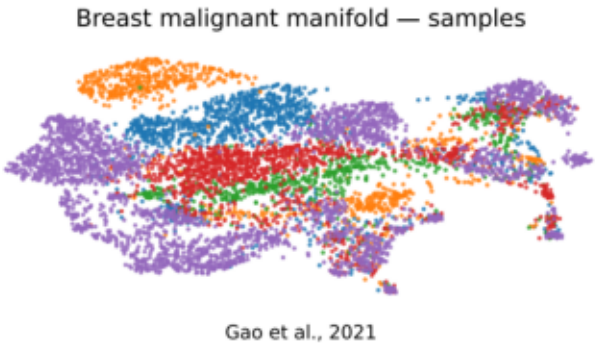
Not evaluated — Gate 1 NO-GO stops the chain. Module scores are flat and low-amplitude everywhere (Gast −0.024 / pSom +0.016 / Crest −0.042); no Gast/pSom hub, no arms.

Gate 3 — cross-repression → N/A

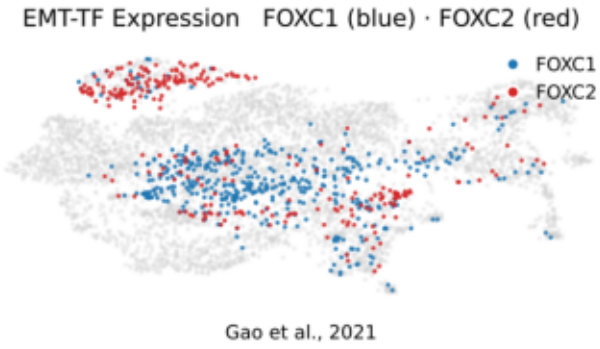
Not evaluated — no trajectory is fit (branching-only gate).

STEP FIGURES

Malignant manifold — samples



EMT-TF expression



Dataset.
Breast cancer (Gao et al., 2021), 10x — 6,411 malignant cells across 5 breast tumours (DCIS1, TNBC1–5) pooled from the /Breast and /Breast_and_thyroid leaves (inner-join 22,164 genes); 5 anaplastic-thyroid (ATC) samples dropped as a different cancer lineage.

Verdict.
The embryonic-program pattern does NOT reproduce. The breast malignant compartment is inter-patient CNV/clonal heterogeneity — a per-patient archipelago no integration strength resolves into one continuous, sample-mixed manifold — so no cell-state trajectory is fit (Gate 1 NO-GO). The only cross-patient axis is proliferation. This corrects prior attempts that forced a curve onto the over-integrated blob.

Parameter choices

Scope	breast-only; 5 ATC samples dropped (different lineage)	**Sample threshold**	≥347 cells (gap 618→195, 3.17×); drops TNBC3 → 5 samples
Embedding	2,663 overdispersed; PCA n_pcs=30 (robust 20/30/40)	**Integration**	Harmony on sample, θ swept 0/2/5/8/10/20; θ=10 plateau
Gate-1 read	per-neighbourhood iLISI / kNN / self-enrichment (not a global scalar)	**EMT-TFs shown**	FOXC1 (28.5%) · FOXC2 (16.5%); descriptive overlay (no pseudotime)